

Grade 7 Accelerated
Unit 1
Lesson 4 Homework

Name _____
Date _____
Period _____

Frankie solved the equation $6y + 2.5 = 46$. Her work is shown for questions 1-2.

$$\begin{aligned}6y + 2.5 - 2.5 &= 46 - 2.5 \\6y &= 43.5 \\6y \div 6 &= 43.5 \div 6 \\y &= 7.25\end{aligned}$$

1. Use substitution to verify if the answer is correct.
2. If correct, describe the properties of equality applied. If incorrect, describe the error(s) made.

Alishia solved the equation $7x - 4\frac{1}{4} = 44.75$. Her work is shown for questions 3-4.

$$\begin{aligned}7x - 4\frac{1}{4} - 4\frac{1}{4} &= 44.75 - 4\frac{1}{4} \\7x &= 40.25 \\7x \div 7 &= 40.25 \div 7 \\x &= 5.75\end{aligned}$$

3. Use substitution to verify if the answer is correct.
4. If correct, describe the properties of equality applied. If incorrect, describe the error(s) made.
5. You pay \$55 for a gym membership, along with an additional monthly fee. Your cost for the year is \$235. The equation $\$55 + 12t = \235 can be used to represent the situation. Solve and then explain what the solution means in this context.

6. You and your best friend go kayaking. The cost to rent a kayak is \$24 for the first 3 hours, and \$4.50 for each additional hour. You spent \$42 for the day. The equation $\$24 + \$4.50m = \$42$ can be used to represent the situation. Solve and then explain what the solution means in this context.
7. Solve $8 + \frac{z}{4} = 23$ using the properties of equality. Then, use substitution to verify your solution is correct.
8. Solve $5t - 17.2 = 16.3$ using the properties of equality. Then, use substitution to verify your solution is correct.
9. Solve $\frac{s}{3} - 0.6 = 1.2$ using the properties of equality. Then, use substitution to verify your solution is correct.
10. Solve $2.4a + 8 = 27.2$ using the properties of equality. Then, use substitution to verify your solution is correct.